



FACULTY OF ENGINEERING AND TECHNOLOGY DEPARTMENT OF
ELECTRICAL AND COMPUTER ENGINEERING

COMMUNICATION SYSTEMS ENEE3309

COURSE PROJECT

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Section: 4

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1. FIND ***a0, a1, a2, a3, b1, b2, and b3*** (YOU CAN USE MATLAB OR ANY OTHER CODE TO FIND NUMERICAL VALUES OF THE COEFFICIENTS)

We can find the coefficient from the equations below:

$$a_0 = \frac{1}{T} \int_{t_0}^{(t_0+T)} x(t) dt$$

$$a_n = \frac{2}{T} \int_{t_0}^{(t_0+T)} x(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{t_0}^{(t_0+T)} x(t) \sin n\omega_0 t dt$$

The code :

```

ayaProject.m x +
1      syms t n;
2      T = 0.1;
3      w =2*pi*(1/T);
4      A = 1;
5      n = 1:3;
6      x(t) = (-2*A* t*(1/T) + 2*A);
7      a0 = (1/T)*int (A,t,0, (T/2)) + (1/T)*int (x(t), t, (T/2),T)
8      an = (2/T)*int (A*cos(n*w*t),t,0, (T/2)) + (2/T)*int(x(t)*cos(n*w*t),t, (T/2),T)
9      bn = (2/T)*int(A*sin(n*w*t),t,0, (T/2)) + (2/T)*int(x(t)*sin(n*w*t),t, (T/2),T)
10     |

```

The output :

$$b1 = 1/\pi \quad b2 = 1/(2*\pi) \quad b3 = 1/(3*\pi)$$

```
>> ayaProject
```

```
a0 =
```

```
3/4
```

```
an =
```

```
[-2/pi^2, 0, -2/(9*pi^2)]
```

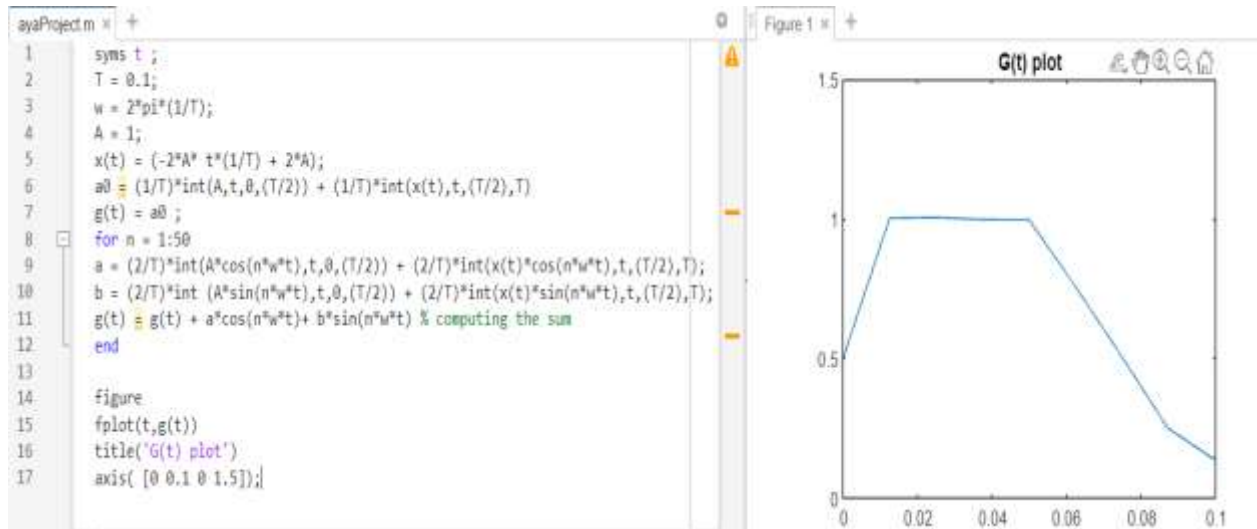
```
bn =
```

```
[1/pi, 1/(2*pi), 1/(3*pi)]
```

$$a0=0.75 \quad a1 = -2/\pi^2 \quad a2=0 \quad a3 = -2/(9*\pi^2)$$

2. USE MATLAB TO PLOT $g(t)$ AND $ga(t)$ FOR $K = 3$, ON THE SAME FIGURE FOR ONE CYCLE OF $g(t)$.

- Plot $g(t)$:

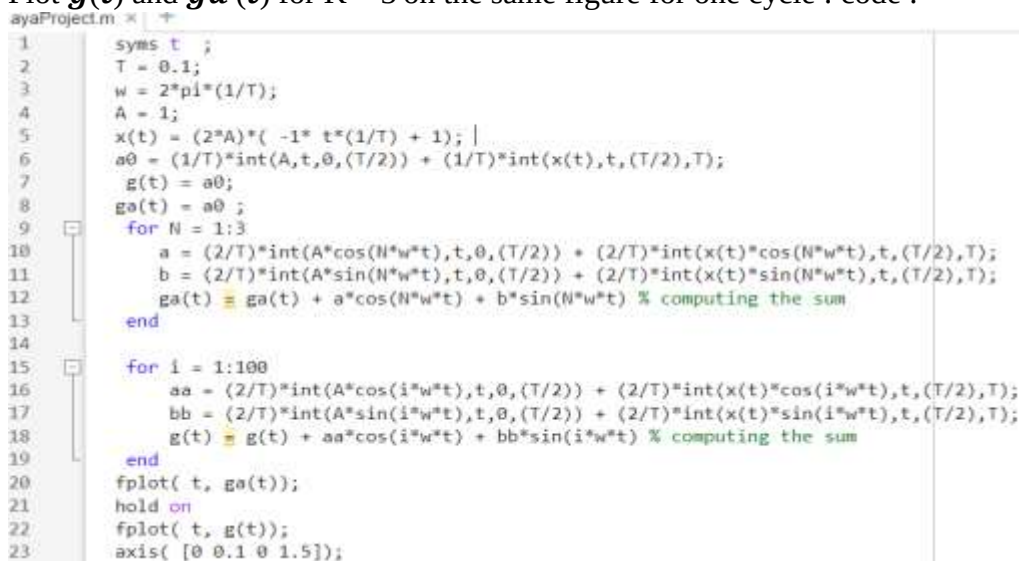


This signal is close to the actual signal since the distortion is small because we take the first 50 terms, and if we take n larger, we approach to the actual signal.

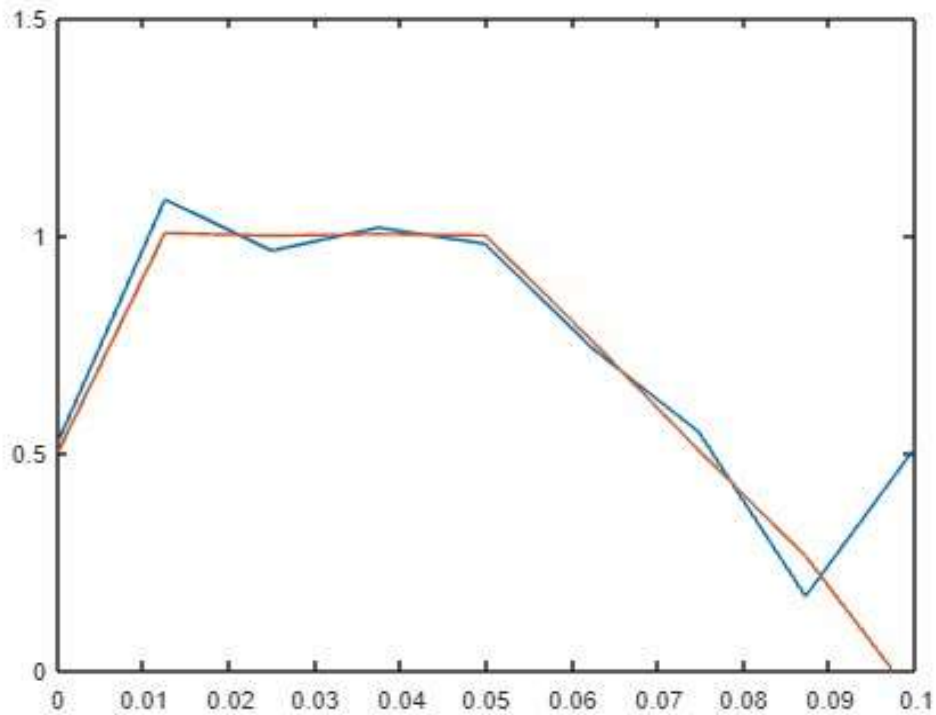
As we know the distortion refers to any unwanted change to a signal as it travels through the system. It can include changes in the amplitude, phase, or frequency of the signal.

Distortion can occur due to a variety of factors such as non-linearities in the system, transmission medium properties, and external interference.

- Plot $g(t)$ and $ga(t)$ for $K = 3$ on the same figure for one cycle : code :



The output :



As the number of terms ($k = 3$) of the trigonometric fourier series, the output signal $g_a(t)$ will try to take the form of the actual signal, but the distortion is high, so the result shown above is disorganized.

3. THE MEAN SQUARE ERROR BETWEEN $g(t)$ AND $g_a(t)$ IS DEFINED AS

$$MSE = \frac{1}{T_0} \left(\int_0^{T_0} (g(t) - g_a(t))^2 dt \right)$$

Code :

```

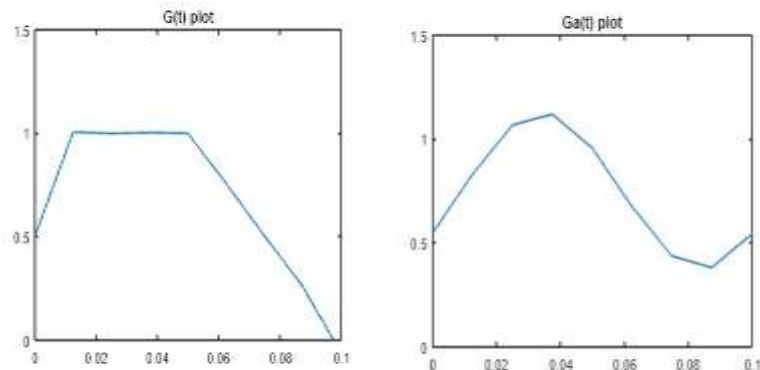
ayaProject.m x +
1      syms t;
2      T = 0.1;
3      W = 2*pi*(1/T);
4      A = 1;
5      x(t)= (-2*A* t*(1/T) + 2*A);
6      a0 = (1/T)*int(A,t,0,(T/2)) + (1/T)*int(x(t),t,(T/2),T);
7      g(t) = a0 ;
8      ga(t) = a0 ;
9
10     for i = 1:110
11         aa = (2/T) *int (A*cos(i*w*t),t,8,(T/2)) + (2/T)*int(x(t)*cos(i*w*t),t,(T/2),T);
12         bb = (2/T)*int (A*sin(i*w*t),t,8,(T/2)) + (2/T)*int(x(t)*sin(i*w*t),t,(T/2),T);
13         g(t) = g(t) + aa*cos(i*w*t)+ bb*sin(i*w*t); % computing the sum
14     end
15
16     for v = 1:3
17         a = (2/T)*int(A*cos(w*t),t,8,(T/2)) + (2/T)*int(x(t)*cos(w*t),t,(T/2),T);
18         b = (2/T)*int(A*sin(w*t),t,8,(T/2)) + (2/T)*int(x(t)*sin(w*t),t,(T/2),T);
19         ga(t)=ga(t) + a*cos(w*t)+ b*sin(w*t);
20         mse = (1/T)*int( ((g(t)-ga(t)).^2),t,0,T)
21     end
22
23     figure
24     fplot(t, ga(t))
25     title('Ga(t) plot')
26     axis( [0 0.1 0 1.5]);
27
28     figure
29     fplot(t, g(t))
30     title('G(t) plot')
31     axis( [0 0.1 0 1.5]);

```

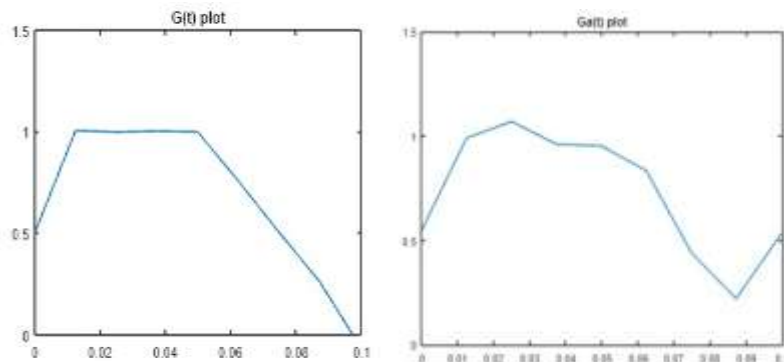
K	MSE
1	$ \begin{aligned} & (10*((6173225496986568425226399518449307155404079145479636181067124800983001 \\ & 44925121901*\pi^2)/1944372288868762061179315953345246288323951167991492483565 \\ & 4407094110359723304960000 + \\ & 1879269953422395197523571117296987757175143181548516219881065001296034104460 \\ & 4367313840728104394293906569895842851281946639435038860990804557049746694403 \\ & 07/6401714272324247621000477867888674604846746180771643224570914312481846409 \\ & 5262577621342789189737929134567295175295763754937546763403284476443520210572 \\ & 4609375))/\pi^4 \end{aligned} $

2	<pre> mse = (10*((3742760135900615848752254576767749294999140185490270576610323914219206 48383809901*pi^2)/1944372288868762061179315953345246288323951167991492483565 4407094110359723304960000 + 1879269953422395197523571117296987757175143181548516219881065001296034104460 4367313840728104394293906569895842851281946639435038860990804557049746694403 07/6401714272324247621000477867888674604846746180771643224570914312481846409 5262577621342789189737929134567295175295763754937546763403284476443520210572 4609375))/pi^4 </pre>
3	<pre> mse = (10*((2662553308751303592541523491575945801485833981050552530185079075657519 83254337901*pi^2)/1944372288868762061179315953345246288323951167991492483565 4407094110359723304960000 + 2985997627250501059185148536207718053611317788888512261598515908066893119848 1753085709036131009780708742674045066489207289296651811674770401311761826843 2/64017142723242476210004778678886746048467461807716432245709143124818464095 2625776213427891897379291345672951752957637549375467634032844764435202105724 609375))/pi^4 </pre>

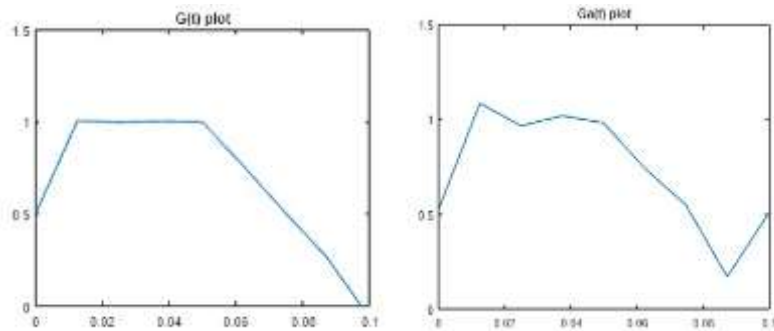
When $k = 1$, the error will be huge because the distortion in this case for $g_a(t)$ is so big and the distortion in $g(t)$ is smaller than $g_a(t)$, and this is shown in the figures below:



When $k = 2$, the error will be huge because the distortion in this case for $g_a(t)$ is so big and the distortion in $g(t)$ is smaller than $g_a(t)$, and this is shown in the figures below:



When $k = 3$, the error will be huge but less than the other cases, because the distortion in this case for $ga(t)$ is big and the distortion in $g(t)$ is smaller than $ga(t)$, and this is shown in the figures below:



4. IF $ga(t)$ (WHEN $K = 3$) IS MULTIPLIED BY THE CARRIER

$$c(t) = 10 \cos 2\pi(200)t$$

FOLLOWED BY AN IDEAL BANDPASS FILTER TO GENERATE THE SINGLE SIDEBAND SIGNAL $s(t)$, FIND $S(T)$ AND ITS SPECTRUM .

Code :

```
fs = 1000; % Sample rate

f0 = 2*pi*1000; % Base frequency
t = linspace(-300, 300, fs); % Time vector
f = linspace(-300, 300, fs); % freq vector
ga = sin(20*pi*t)/pi - (2*cos(60*pi*t))/(9*pi^2) - (2*cos(20*pi*t))/pi^2 +
sin(40*pi*t)/(2*pi) + sin(60*pi*t)/(3*pi) + 3/4; % message signal

% Modulate the signal with a carrier frequency
fc = 2*pi*200; % Carrier frequency
y = ga.*cos(fc.*t); % AM modulated signal

figure
plot(t, y)
title('s(t) before bandpass filter')
yf = fft(y)

figure
plot(f, abs(yf))
title('spectrum of s(t) before bandpass filter')

% Design a bandpass filter to take the LSB
%since the frequencies of s(t) before filter is 170,190,200,210,230
low = 205; % Lower cutoff frequency
```

```

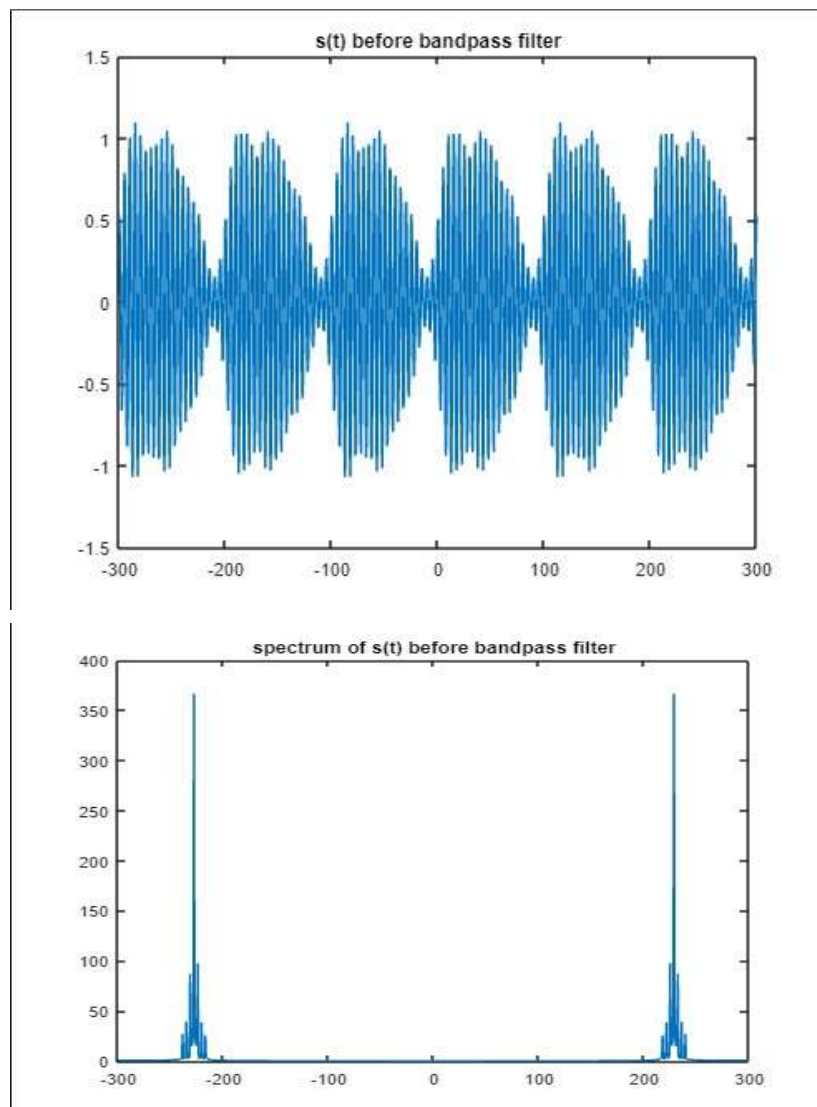
high = 235; % Upper cutoff frequency
fc1 = low/(fs/2);
fc2 = high/(fs/2);
[b,a] = butter(5,[fc1 fc2],'bandpass');

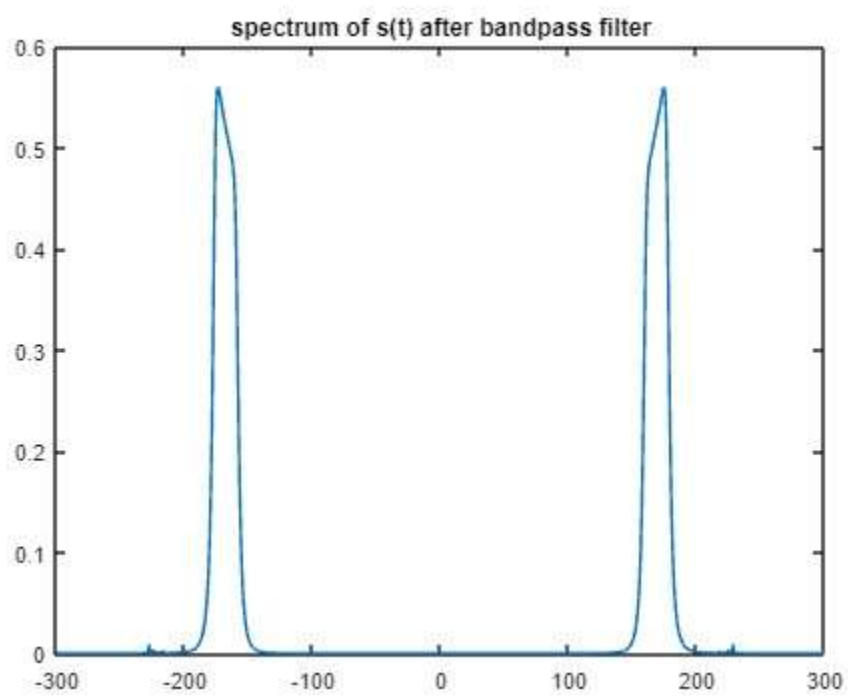
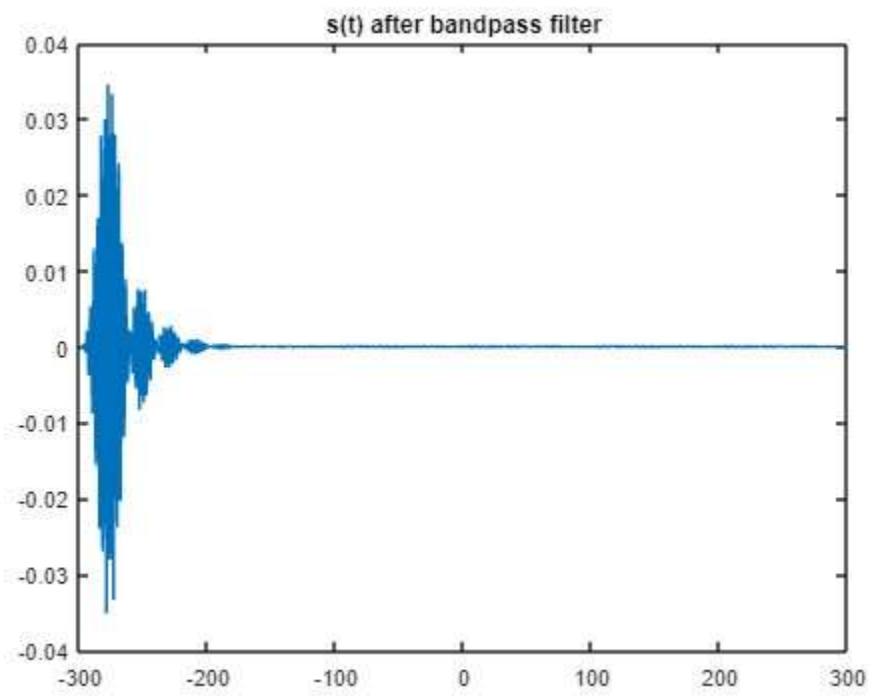
% Filter the modulated signal to remove one sideband s = filter(b,a,y);
figure
plot(t, s)
title('s(t) after bandpass filter')

sf = fft(s);
figure
plot(f, abs(sf))
title('spectrum of s(t) after bandpass filter')

```

Output :





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$$g_a(t) = \sum_{n=1}^3 a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) + a_0$$

we know that $a_0 = \frac{3}{4}$, $a_1 = -\frac{2}{\pi^2}$, $a_2 = 0$, $a_3 = \frac{2}{9\pi^2}$
 $b_1 = \frac{1}{\pi}$, $b_2 = \frac{1}{2\pi}$, $b_3 = \frac{1}{3\pi}$

$$\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{0.1} = 20\pi$$

$$s(t) = g_a(t) \cdot c(t), \quad c(t) = 10 \cos(2\pi(200)t)$$

$$s(t) = \left[a_0 + \sum_{n=1}^3 (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)) \right] \cdot 10 \cos(2\pi(200)t)$$

$$= \frac{30}{4} \cos(2\pi(200)t) + \sum_{n=1}^3 10 a_n \cos(n\omega_0 t) \cos(2\pi(200)t) \\ + \sum_{n=1}^3 10 b_n \sin(n\omega_0 t) \cos(2\pi(200)t)$$

$$= 10 a_1 \cos(n\omega_0 t) \cos(2\pi(200)t) + 10 a_2 \overset{\text{zero}}{\cancel{\cos(n\omega_0 t) \cos(2\pi(200)t)}} \\ + 10 a_3 \cos(n\omega_0 t) \cos(2\pi(200)t)$$

$$= 10 \cdot -\frac{2}{\pi^2} \cos(\omega_0 t) \cos(2\pi(200)t) + 10 \left(-\frac{2}{9\pi^2} \right) \cos(3\omega_0 t) \cos(2\pi(200)t)$$

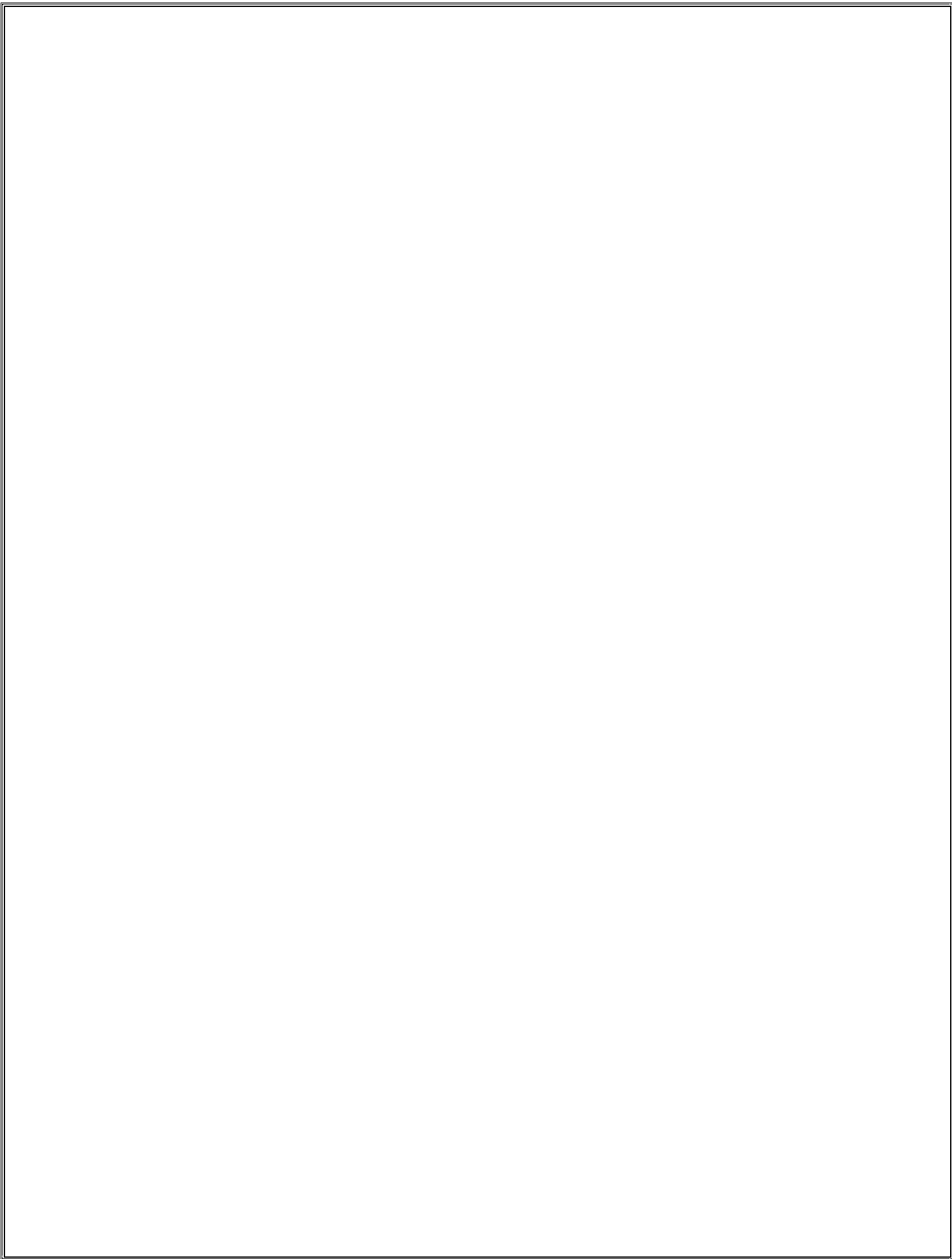
$$= -\frac{20}{\pi^2} \cos(20\pi t) \cos(2\pi(200)t) + \frac{-20}{9\pi^2} \cos(3(2\pi(200)t) \cos(2\pi(200)t)$$

$$= -\frac{20}{\pi^2} \cos(2\pi(20+200)t) \cdot \frac{1}{2} - \frac{20}{9\pi^2} \cdot \frac{1}{2} \cos(2\pi t(30+200))$$

$$= -\frac{10}{\pi^2} \cos(2\pi t(200+10)) - \frac{10}{9\pi^2} \cos(2\pi t(200+30))$$

$$= \left[-\frac{10}{\pi^2} \cos(2\pi t(210)) - \frac{10}{9\pi^2} \cos(2\pi t(230)) - \frac{10}{\pi^2} \cos(2\pi t(190)) \right.$$

$$\left. - \frac{10}{9\pi^2} \cos(2\pi t(170)) \right]$$



* Then find 3th term:-

$$\sum_{n=1}^3 10 b_n \sin \cos(2\pi t(200))$$

$$= \frac{10}{\pi} \sin(\omega_0 t) \cos(2\pi t(200)) + \frac{10}{2\pi} \sin(2\omega_0 t) \cos(2\pi t(200)) \\ + \frac{10}{3\pi} \sin(3\omega_0 t) \cos(2\pi t(200))$$

$$* \sin \alpha \cos \beta = \frac{1}{2} \sin(\alpha - \beta) + \frac{1}{2} \sin(\alpha + \beta)$$

$$= \frac{5}{\pi} \sin(2\pi t(210)) - \frac{5}{\pi} \sin(2\pi t(190)) + \frac{5}{2\pi} \sin(2\pi t(220)) \\ - \frac{5}{2\pi} \sin(2\pi t(180)) + \frac{5}{3\pi} \sin(2\pi t(230)) - \frac{5}{3\pi} \sin(2\pi t(170))$$

* we need it in Cos :-

$$= \left[\frac{5}{\pi} \cos(2\pi t(210) - \frac{\pi}{2}) + \frac{5}{\pi} \cos(2\pi t(190) + \frac{\pi}{2}) \right. \\ \left. + \frac{5}{2\pi} \cos(2\pi t(220) - \frac{\pi}{2}) + \frac{5}{2\pi} \cos(2\pi t(180) + \frac{\pi}{2}) \right. \\ \left. + \frac{5}{3\pi} \cos(2\pi t(230) - \frac{\pi}{2}) + \frac{5}{3\pi} \cos(2\pi t(170) + \frac{\pi}{2}) \right]$$

$$\Rightarrow S(t) = \cancel{\frac{30}{4}} \frac{30}{4} \cos(2\pi t(200)) - \frac{10}{\pi^2} \cos(2\pi t(210)) - \frac{10}{9\pi^2} \cos(2\pi t(230)) \\ - \frac{10}{\pi^2} \cos(2\pi t(190)) - \frac{10}{9\pi^2} \cos(2\pi t(170)) + \frac{5}{\pi} \cos(2\pi t(210) - \frac{\pi}{2}) \\ + \frac{5}{\pi} \cos(2\pi t(190) + \frac{\pi}{2}) + \frac{5}{2\pi} \cos(2\pi t(220) - \frac{\pi}{2}) + \frac{5}{2\pi} \cos(2\pi t(180) + \frac{\pi}{2}) \\ + \frac{5}{3\pi} \cos(2\pi t(230) - \frac{\pi}{2}) + \frac{5}{3\pi} \cos(2\pi t(170) + \frac{\pi}{2})$$